# Kooshan Amini

Machine Learning Research Scientist

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Ph.D. expected Dec 2027 · Available for Summer 2027 internships · No visa sponsorship required

## SUMMARY

Machine learning researcher and Ph.D. candidate at Rice University adapting vision-language and depth foundation models to real-world imagery under domain shift, and building probabilistic and physics-informed models with uncertainty quantification. First-author peer-reviewed papers, released model weights and datasets, and hands-on engineering from data curation to large-scale inference.

## TECHNICAL SKILLS

**Machine learning:** vision-language and foundation models, semantic segmentation, monocular depth, physics-informed and monotonic networks, Bayesian inference, Gaussian processes, uncertainty quantification

**Programming & frameworks:** Python (expert), SQL, C++, MATLAB; PyTorch, PyTorch Lightning, Hugging Face Transformers, vLLM, scikit-learn, OpenCV, Weights & Biases

**Modeling & computing:** Monte Carlo simulation, random fields, multi-objective optimization, network analysis, interpretability (ALE/PDP), GPU computing (CuPy), Linux/HPC (TACC), Git, geospatial data (GDAL, LiDAR)

## RESEARCH & ENGINEERING EXPERIENCE

**Graduate Research Assistant**, Rice University — Aug 2022 – Present

Co-advised by Engineering and Computer Science faculty; projects funded by NSF, NASA, NIST, and Harris County

### Vision-language segmentation under domain shift — NSF

- Adapted a frozen CLIP-based segmentation model with visual-prompt engineering and text–image prompt mixing, updating under 1% of its parameters.
- Designed a cross-storm split and multi-annotator consensus labels; the model generalized to an unseen hurricane (Dice 0.86 vs. 0.54 zero-shot), ahead of DeepLabV3 and U-Net, while SAM and Grounded-SAM failed zero-shot.
- Reported a useful negative result: single-region training generalized better than pooled multi-region data.

### Monocular 3D estimation from single aerial images — NSF

- Built a lightweight head on frozen Depth Anything V2 conditioned on segmentation masks, trained on LiDAR-fused targets with a Gaussian-likelihood loss that penalizes under-prediction.
- Closely matched UAV surveys across 10 regions and 5 hurricanes (Spearman $\rho = 0.87$), while standard engineering methods over-predicted volume 3–5×; regional calibration reduced error further.

### Vision-language model evaluation on satellite imagery — NASA ROSES

- Benchmarked four open VLMs on road and bridge damage: reasoning mode, visual prompting, and image co-registration more than doubled damage recall; found that output-length caps were truncating model reasoning.
- Designed a blind, double-labeled benchmark with spatially blocked sampling; ran inference on 30,000+ segments in about three days on two GPUs.

### Physics-informed learning and uncertainty quantification — NSF

- Built a monotonicity-constrained network after Gaussian processes, gradient boosting, and unconstrained networks produced implausible extrapolations.
- Coupled it with sea-level-rise and land-cover scenarios in 9,000 Monte Carlo runs: 11–14% more debris by 2040.
- Random-field and Monte Carlo network models showed debris raises loss of emergency-medical access from 2% to 17%.

### Bayesian data fusion for flood situational awareness — NSF OpenSafe.AI

- Fused crowd reports, traffic cameras, and gauges with ensemble flood forecasts, improving held-out flood detection in every forecast cycle tested; designed the multi-team system architecture and interface standards.

**Decision science for community resilience** — NIST Center of Excellence

- Multi-objective optimization over an integrated hazard–damage–recovery–economics model found a $200M plan that cuts expected losses by $192M, and showed that efficiency does not guarantee equity.

**Risk & Resilience Engineering Intern**, Arup — Jun – Aug 2025

- Delivered flood-risk and transportation-resilience analyses for industry and public-sector clients; full-time offer.

## OPEN SOURCE & DATASETS

- **CLIPSeg-debris:** released model weights, a consensus-labeled aerial-image dataset, and code.
- **Research code:** hurricane debris projection framework and coastal resilience decision-support toolkit, public on GitHub.
- **IN-CORE (NCSA/NIST):** contributed a power-network recovery module (1,400+ lines) via a merged pull request.

## SELECTED PUBLICATIONS

[1] **Amini, K.**, Meads, M. M., Padgett, J. E., et al. (2026). Evolving threats: Understanding the coupled impact of sea level and land-cover changes on hurricane-induced debris risk. *ASCE Natural Hazards Review*, in press.

[2] **Amini, K.**, Padgett, J. E., & Balakrishnan, G. (2026). Rapid debris-volume estimation from post-hurricane aerial imagery. *arXiv:2608.17165*; under review at *ISPRS Journal of Photogrammetry and Remote Sensing*.

[3] **Amini, K.**, Palit, T., Beck, A. L., et al. (2026). Integrated decision-support framework for enhancing coastal community resilience against hurricane-induced hazards. *International Journal of Disaster Risk Reduction*, 140, 106136.

[4] **Amini, K.**, Liu, Y., Padgett, J. E., Balakrishnan, G., & Veeraraghavan, A. (2025). Debris segmentation using post-hurricane aerial imagery. *Computer-Aided Civil and Infrastructure Engineering*, 40(25), 4116–4131.

[5] **Amini, K.**, & Padgett, J. E. (2023). Probabilistic risk assessment of hurricane-induced debris impacts on coastal transportation infrastructure. *Reliability Engineering & System Safety*, 240, 109579.

[6] Nofal, O. M., **Amini, K.**, Padgett, J. E., van de Lindt, J. W., et al. (2023). Multi-hazard socio-physical resilience assessment of hurricane-induced hazards on coastal communities. *Resilient Cities and Structures*, 2(2), 67–81.

Also: conference papers at ICOSSAR 2025 and ICASP14 2023; 130+ citations (Google Scholar).

## EARLIER RESEARCH

**Research Assistant & Software Developer**, Sharif University of Technology — 2018 – 2022

- Probabilistic models of earthquake-induced social impacts; developer on Rtx, a reliability and risk platform.

## EDUCATION

**Ph.D. in Engineering**, Rice University · GPA 3.96/4.00 — 2022 – Expected Dec 2027

**M.Sc. in Engineering**, Sharif University of Technology · GPA 3.42/4.00 — 2017 – 2020

**B.Sc. in Engineering**, University of Guilan · GPA 3.88/4.00, ranked 3rd of 80+ — 2013 – 2017

## AWARDS & HONORS

- Selected for the first IN-CORE Contributor Cohort (NCSA and the NIST Center of Excellence), 2024; travel award, NHERI Computational Symposium, 2023.
- Top 0.2% nationally in the graduate engineering entrance exam; 16th nationally in the engineering Scientific Olympiad.

## LEADERSHIP & SERVICE

- **Invited talks:** NDPTC webinar (2026), NSF NHERI DesignSafe webinar (2025), and NHERI SimCenter roundtable (2024, the only graduate-student speaker); conference talks at AEI/IRD, ICOSSAR, ASCE EMI, ICASP14, and NHERI.
- **Mentoring & teaching:** D2K Research Mentoring Fellow for data-science capstone teams (2026); teaching assistant for a graduate resilience course (2025); Vice President, Graduate Student Association (2022–2024).
- **Service:** journal reviewer for *Climatic Change* and *Natural Hazards and Earth System Sciences*; expert novelty assessor for the UK Metascience Novelty Indicators Challenge (UKRI).